Piyush Maji

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.50
Sugandha High School	Class XII (West Bengal Council of Higher Secondary Education)	2021	75.00%
Jadabpur High School	Class X (West Bengal Board of Secondary Education)	2019	86.00%

RESEARCH

Bayesian Optimization for Accelerator Parameter Tuning | Belle II Experiment, KEK, Japan

Python - Gaussian Processes - SHAP | Advisor: Prof. Sourav Sandilya, Belle II Collaboration

- Addressing multi-parameter accelerator tuning under noisy, black-box objectives by implementing a **Bayesian Optimization** framework, motivated by injection-tuning methods at **SuperKEKB (KEK)**.
- Engineered a **Gaussian Process**-based optimizer with **EI/UCB acquisition functions**, **segmented region search**, step-by-step parameter constraints, and **proximal biasing** to navigate noisy, non-convex parameter spaces.
- Benchmarked against **random/grid search & Nelder-Mead** on simulated objectives, used **SHAP** for per-parameter attribution.

PROJECTS

FraudNet | AI-Powered Fraud Investigation Platform

Self-Initiated

- Engineered an **end-to-end graph-based fraud detection system** on **203K+ Bitcoin transactions and 234K+ transaction edges**, benchmarking GraphSAGE and GAT against Logistic Regression, Random Forest, and XGBoost; XGBoost achieved **0.954 F1 and 0.987 PR-AUC** on illicit transactions.
- Built an **agentic investigation workflow with LangGraph** chaining prediction -> SHAP/GNNExplainer -> graph-neighborhood analysis -> RAG via ChromaDB -> Gemini API report generation; deployed a **Dockerized FastAPI REST service** with prediction, node-analysis, and investigation endpoints, supported by pytest and GitHub Actions CI.

Retail Demand Forecasting & Inventory Optimization

Self-Initiated

- Built a **config-driven demand forecasting pipeline** using a global **LightGBM** model with Tweedie objective and early stopping across store × SKU series, benchmarked against SARIMA and Prophet using rolling-origin cross-validation.
- Engineered **20+ leakage-safe temporal features**, including shifted lags, rolling statistics, calendar seasonality, and promotion/holiday indicators; evaluated forecasts using WMAPE, RMSE, bias, and **stockout/overstock cost impact**.
- Implemented **hierarchical forecast reconciliation** using Bottom-Up and MinT trace minimization to enforce consistency across SKU -> store -> region forecasts.

Spam Classification from Scratch

Self-Initiated

- Implemented an **end-to-end text-classification pipeline from scratch in NumPy**, developing Multinomial Naive Bayes and Logistic Regression with custom BoW/TF-IDF, MLE, Laplace smoothing, and gradient descent; achieved **94.9% F1 and 0.982 AUC** on 5,572 SMS messages.

- Designed a **leakage-safe ML pipeline** with train-only feature engineering, L2 regularization, numerically stable log-space inference and custom ROC/PR evaluation, with no reliance on scikit-learn.

SKILLS

Languages: Python, SQL, bash

AI/ML: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, LightGBM, GNNs, Explainable AI, RAG

GenAI & Agents: LangGraph, Gemini API, ChromaDB, Semantic Retrieval, GNNExplainer, SHAP

Engineering & Cloud: FastAPI, Docker, AWS EC2, REST APIs, Git, GitHub Actions, pytest, CI/CD, DSA

RELEVANT COURSES

Machine Learning for Physical Sciences - **Machine Learning for High Energy Physics** - **Data Science Analysis** -**Computational Physics** - **Linear Algebra** - **Statistical Mechanics**

POSITIONS OF RESPONSIBILITY

Inbound & Logistics, International Relations Cell, IIT Hyderabad

- Hosted and coordinated visits for students from Japanese universities, including **Kyoto, UTokyo, Shimane, Nagoya, Ehime, Osaka, Tohoku, and Hokkaido University**.
- Facilitated interactions and visits involving corporate officials from **DNP, Mitsubishi**, and other organizations.

Placement Coordinator, Office of Career Services, IIT Hyderabad

- Engaged prospective recruiters and facilitated their participation in campus recruitment, connecting companies with students throughout the placement process.
- Hosted and coordinated major recruitment events, including **Japan Career Day 2026** and **Career Fair 2026**.

Coordinator, Extra Mural Lectures (EML), IIT Hyderabad

- Led visual design and event branding, managed **100K+ budgets and logistics**, and hosted distinguished speakers, including **NASA Axiom Mission Pilot Shubhanshu Shukla and Ericsson's CTO**.

ACHIEVEMENTS

Accenture Innovation Challenge 2026 -- Selected

Designed a GNN-based digital twin for vehicle assembly-line optimization, modeling stations and workflow dependencies as graph structures to predict bottlenecks and throughput issues.